09-06 第1篇

阅读理解（说明文） Why Do We Have Fingerprints on Our Fingers?

① Hold your fingertip up to a strong light. You will see tiny lines and curves, arranged in patterns of loops, arches and swirls. These are your **fingerprints**. They form when a baby is about three months old inside the mother's body, and they stay almost exactly the same from that day onwards. No two people — not even **identical twins** — share the same pattern. Police have used this fact for more than a hundred years. Yet a simpler question has puzzled scientists longer: **why do our fingertips have these lines at all?**

② For a long time, the most popular answer was: fingerprints help us **grip** things. Smooth hands would slide off a glass of water, a pencil, or a swinging rope; rough patterns on the skin create extra **friction** so that objects do not slip through our fingers. This explanation sounded so reasonable that most textbooks repeated it for years.

③ However, careful experiments in the **21st century** have shown that things are not so simple. In 2009, scientists from the **University of Manchester** used a device that measured the actual friction between a human fingertip and a polished plate of glass. To their surprise, fingertips with their ridges produced about the **same** friction as the same fingers whose ridges had been pressed down flat. In some tests, the ridged fingers even gripped a **little less**, because only the tops of the ridges — not the whole pad of skin — actually touched the surface. So the old "better grip" idea cannot fully explain fingerprints.

④ What, then, do fingerprints really do? Today, most researchers believe fingerprints carry several different jobs. The first and most important may be **helping us feel**. Under each ridge lies a fine set of **sensitive nerves**. As our fingers slide across something, the ridges cause **tiny vibrations** that those nerves read, the way a record needle reads a spinning disc. This helps us tell a very smooth surface from a very fine sand, or a silk scarf from thin cotton, with almost no visual clues.

⑤ A second job may be to help our skin deal with **wet or oily surfaces**. On dry paper, a smooth fingertip grips as well as a ridged one. But on wet glass or a freshly washed plate, the ridges channel water away from the main contact area, a bit like the **grooves** on a car tyre. Water has less space to stay trapped between the skin and the plate, so there is less slipping. In this sense, fingerprints may not give you a huge advantage when holding a **dry** pencil, but they may quietly help you hold a **wet** glass of orange juice without dropping it.

⑥ A third, smaller job is **protection**. The ridges slightly raise the top of the fingertip away from hard surfaces. When you push a heavy door, tiny shocks first hit the ridges, not the softer skin underneath. Over a long life, this may be one reason fingertips wear down less quickly than fully smooth skin would.

⑦ Fingerprint patterns are shaped partly by **genes** and partly by the tiny movements a baby makes while growing. The main pattern — loop, arch or whorl — usually runs in families. Some estimates say about **65%** of people have **loop** patterns, around **30%** have **whorls**, and only about **5%** have pure **arches**. But the fine details of each ridge are shaped by chance, which is why even twins with identical DNA do not share identical prints.

⑧ Sadly, fingerprints are not **indestructible**. Heavy chemicals, long hours of hot washing, or certain illnesses can make the ridges **temporarily less clear**, but the basic pattern usually returns. Only very deep skin damage, reaching the layer below the surface, can remove a print for good. This is one reason people born without fingerprints — a rare condition called "adermatoglyphia" — are considered a very small medical mystery.

⑨ So, next time you press a thumb onto a school sign-in machine, give that small patch of skin a moment of respect. It does not only prove **who** you are to the computer. It also helps you feel the texture of your clothes, hold a wet water bottle, and protect the soft tissue beneath. Your fingerprint is one of the quiet everyday engineers of your daily life.

注释： fingerprint 指纹；ridge 脊；friction 摩擦；groove 凹槽；vibration 振动

根据文章内容，选择最佳答案。

1. When do fingerprints first form on a human being?

A. At birth, during the first minute.
B. When a baby is about three months old inside the mother's body.
C. When a child starts primary school.
D. Only when a teenager becomes 15 years old.

2. What did the 2009 experiment at the University of Manchester show?

A. Fingertips with ridges gave far more friction than pressed-flat fingertips.
B. Fingertips with ridges gave about the same friction as pressed-flat fingertips, and sometimes slightly less.
C. Fingertips without ridges cannot grip anything at all.
D. The scientists could not measure friction at all.

3. How do fingerprints mainly help us feel things, according to paragraph 4?

A. They shine a small light on the surface.
B. They make tiny vibrations as we slide over surfaces, which are read by nerves under the ridges.
C. They make sounds we can hear.
D. They freeze the skin so nerves work faster.

4. Which of the following jobs of fingerprints is NOT mentioned in the passage?

A. Helping us feel fine textures.
B. Channelling water away on wet surfaces, a bit like tyre grooves.
C. Slightly protecting the softer skin underneath from shocks.
D. Allowing us to fly if we flap our hands fast enough.

5. Why, according to the passage, do identical twins still have different fingerprints?

A. They actually share exactly the same fingerprints; the passage is wrong.
B. Their fingers are washed at different times.
C. The fine details of ridges are shaped by chance events during growth, not only by DNA.
D. Only the older twin has fingerprints.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第①段 "They form when a baby is about three months old inside the mother's body"
【解析】胎儿约三个月时指纹形成。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "fingertips with their ridges produced about the same friction ... sometimes gripped a little less"
【解析】实验推翻了"脊多摩擦大"的旧说法。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "the ridges cause tiny vibrations that those nerves read"
【解析】脊纹带来振动被神经读取。A、C、D 均不合事实。故选 B。

4.
【答案】D

【题型】细节判断题（否定型）
【定位】第④—⑥段列出感知、排水、保护三种功能
【解析】让人飞行并非指纹作用。A、B、C 均为原文所述。故选 D。

5.
【答案】C
【题型】细节理解题
【定位】第⑦段 "the fine details of each ridge are shaped by chance, which is why even twins with identical DNA do not share identical prints"
【解析】纹路细节源于发育中的随机变化。A、B、D 均误。故选 C。